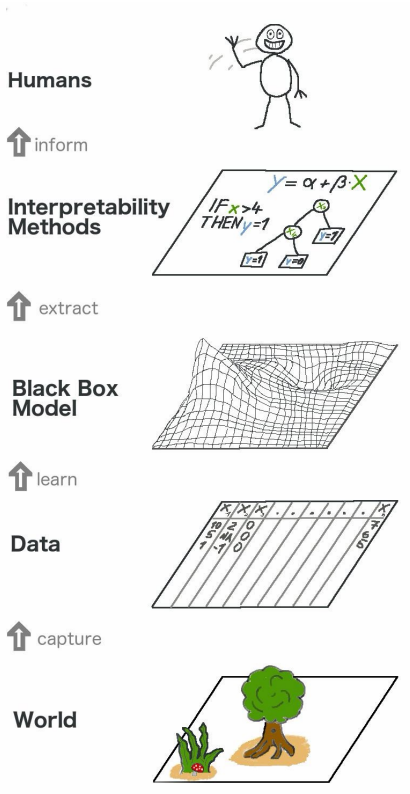
CHAPTER 4 · INTERPRETABLE MACHINE LEARNING

Methods Overview

A roadmap of interpretability methods

Asasira Martin

Source: C. Molnar, Interpretable Machine Learning, 3rd ed. (2025)

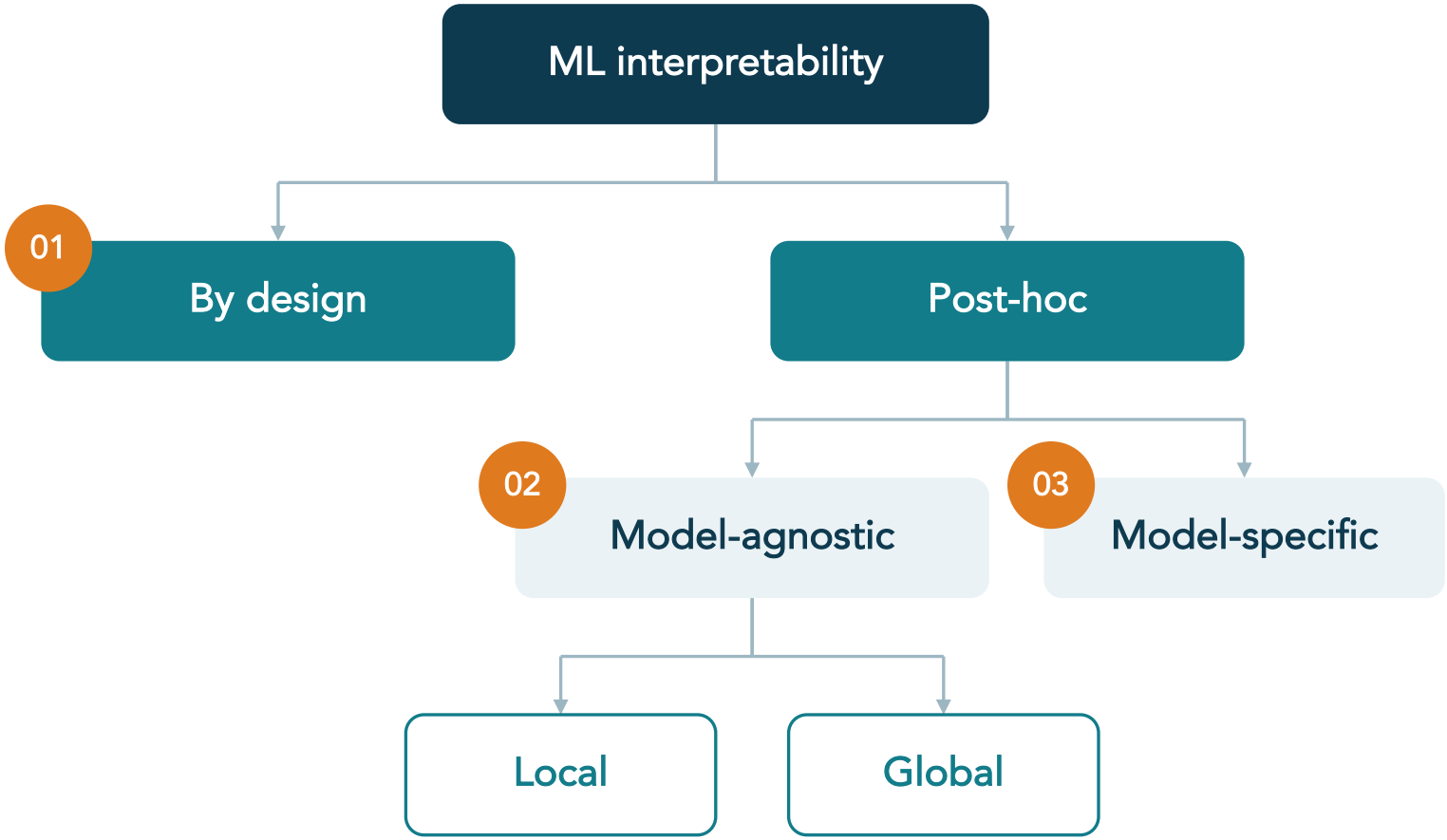


World → data → model → interpretability methods → humans

Figure: Molnar (2025)

ROADMAP

Interpretability splits into two main paths



01 **Interpretability by design**
Interpretable models and their scope

02 **Model-agnostic methods**
SIPA, local and global methods

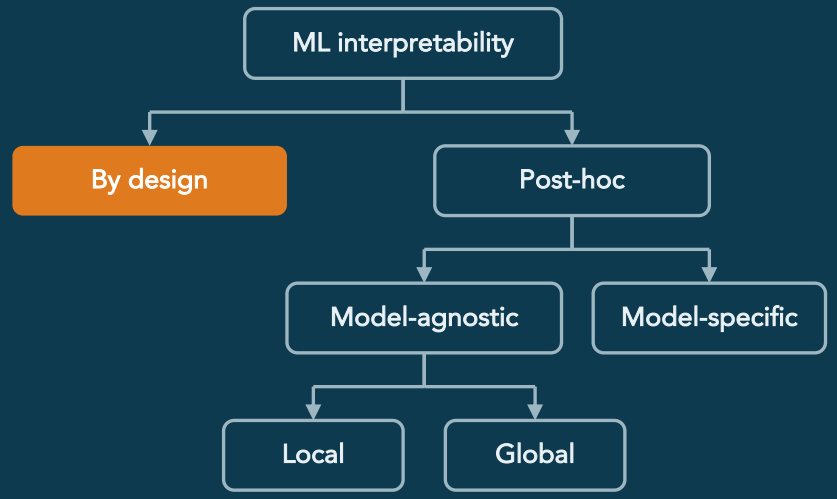
03 **Model-specific methods**
Looking inside neural networks

04 **Synthesis**
Goals and blurry lines

01

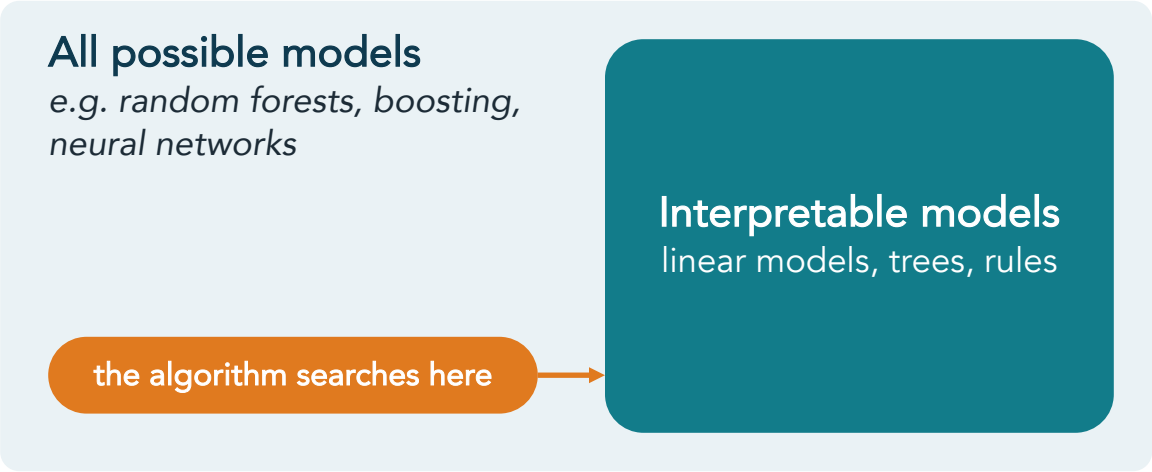
Interpretability by design

Training interpretable models from the start



The algorithm | Scope | Strengths

The learning algorithm searches only interpretable models



Interpretability is decided at the level of the algorithm.

Example: ordinary least squares only finds models that are linear in the features.

Also called: intrinsically or inherently interpretable models.

The six approaches the book covers

Linear regression

Minimizes squared errors

Logistic regression

Linear model for classification

Linear model extensions

Penalties, interactions, non-linear terms

Decision trees

Split the data recursively

Decision rules

If-then rules from data

RuleFit

Tree rules + Lasso

Four more interpretable approaches



ProtoViT

Ma et al. (2024)

Classifies images as a weighted sum of prototypes from the training data.



Interpretable tree ensembles

Yang et al. (2024)

Shallow trees, effects sorted into main effects and interactions, pruning.



Model-based boosting

Bühlmann & Hothorn (2007)

Weighted sum of linear effects, splines, tree stumps and other weak learners.



GAMs with interactions

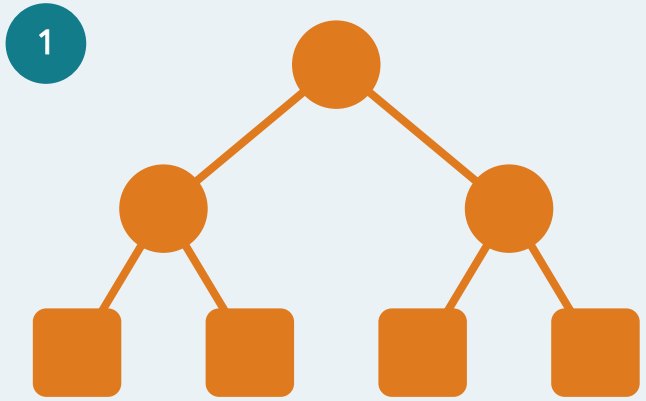
Caruana et al. (2015)

Generalized additive models with automatic interaction detection.

Note: Yang et al.'s tree ensembles mix interpretability by design with post-hoc interpretability.

How interpretable? It depends on the scope

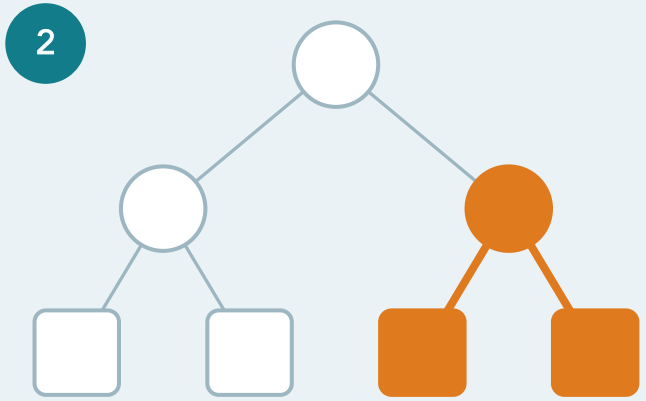
1



The entire model

A small tree or a sparse linear model can be understood as a whole.

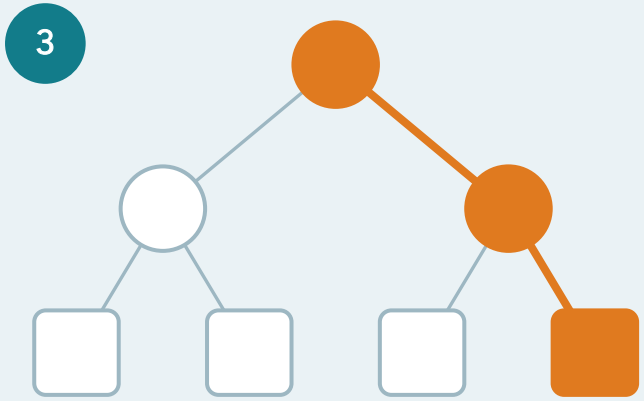
2



Parts of the model

One coefficient or one rule, even if the whole model is too big.

3



Individual predictions

The decision path to one prediction, or its k most similar images.

Tip: for any new interpretability approach, ask whether it explains the entire model, parts of it, or individual predictions.

STRENGTHS

Strong for debugging, but beware Rashomon

Debugging and improving

●●● Strong

Inner workings are visible, so the model is easy to debug.

Justification

●●● Strong

Faithfully explains predictions; easy to check with domain experts.

Insights into the data

●●○ Mixed

Only valid if the model structure reflects the world.

The Rashomon effect

The 1950 Japanese film Rashomon tells four versions of the same murder. Each explains the events equally well, yet they are incompatible.



4 stories, none compatible

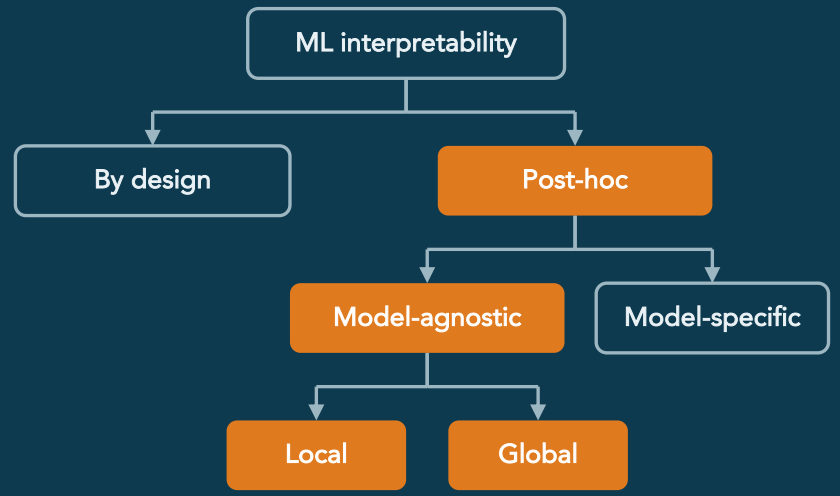
In ML: models with similar performance can tell different stories. Which one should we interpret?

02

Model-agnostic methods

Post-hoc methods that work for any model

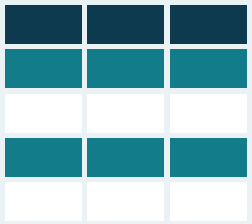
SIPA | Flexibility | Local | Global



Sample, intervene, predict, aggregate

1 Sample

Take a sample of the data

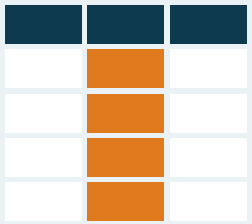


PFI

Sample the test data

2 Intervene

Change the data

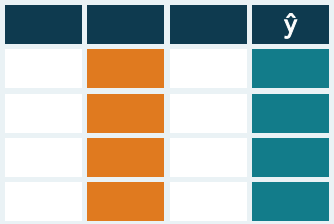


PFI

Shuffle the values of one feature

3 Predict

Predict on the changed data

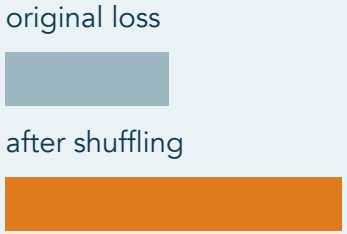


PFI

Get new model predictions

4 Aggregate

Summarize the results



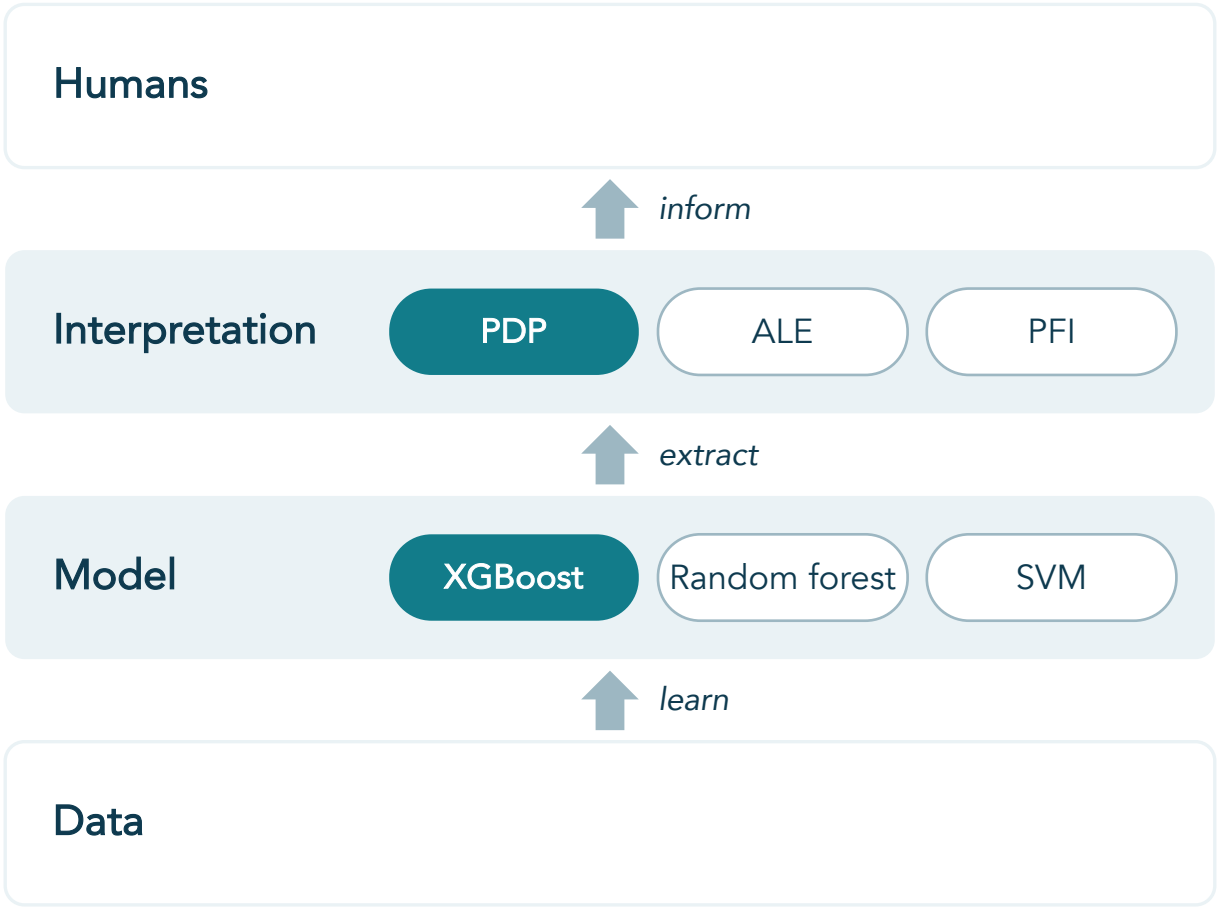
PFI

Compare the new loss with the original loss

Example: permutation feature importance (PFI). Only inputs and outputs are used (Scholbeck et al., 2020).

FLEXIBILITY

Separating explanation from the model brings flexibility



Each grey layer can be swapped without touching the other.

Swap the model

Keep the PDP and replace XGBoost with any other model.

Swap the method

PDP doesn't fit? Use ALE without retraining XGBoost.

Swap the representation

Explain images with patches instead of single pixels.

Compare: with linear regression coefficients, switching to a rule-based model also changes the interpretation tool.

LOCAL METHODS

Local methods explain a single prediction

← one feature at a time

the whole prediction →

Sensitivity

How does the prediction react to one feature?

Ceteris paribus
Change one feature, watch the prediction

ICE curves
The same, for many data points

In between

Which subset of features matters?

Counterfactuals
Smallest change for a desired outcome

Anchors
Values that "anchor" the prediction

Attribution

How is the prediction split among features?

LIME
Local interpretable surrogate model

Shapley values
Fair split among the features

SHAP
Shapley values + global summaries



Debugging
Zoom in on the worst prediction: hard data point, weak model, or wrong label?



Justification
CP and counterfactuals are faithful; SHAP/LIME less so for high stakes (Rudin, 2019).

GLOBAL METHODS

Global methods explain average behavior

Feature effects

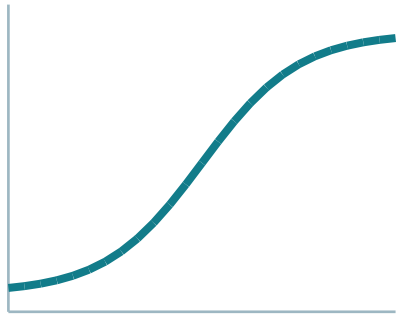
How does the output change with an input?

PDP
partial dependence

ALE
works with correlated features

H-statistic
interaction strength

Decomposition
split into parts



schematic

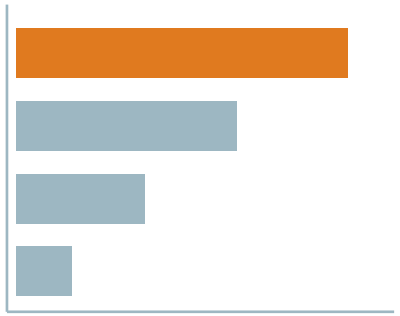
Feature importance

Which features matter most?

PFI
loss increase when permuted

LOFO
drop the feature, retrain

SHAP importance
from Shapley values



schematic

Also: surrogate models, and prototypes and criticisms (representative data points).

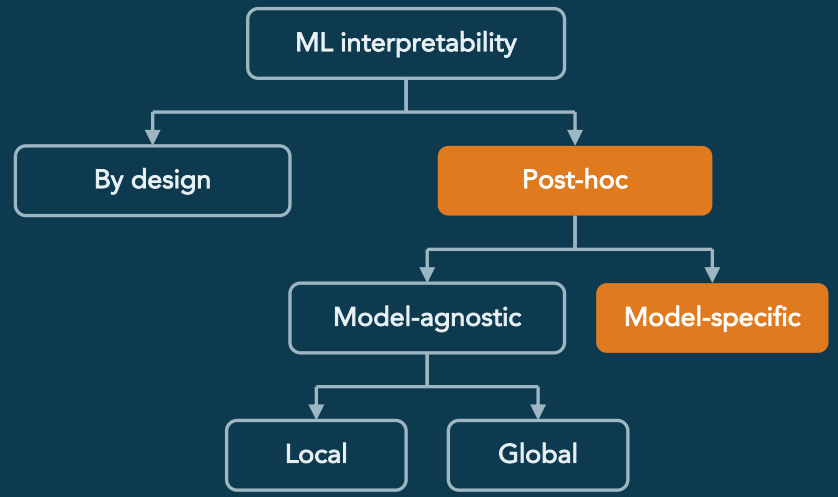
Tip: run global methods on subsets of the data for "group-wise" or "regional" insights.

03

Model-specific methods

Post-hoc methods built for one type of model

Neural networks



Model-specific methods look inside neural networks

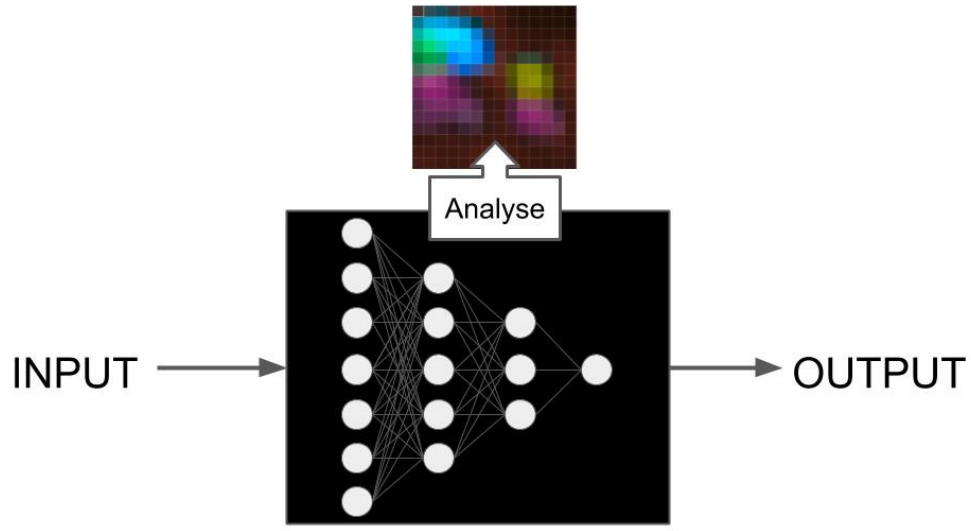


Figure: Molnar (2025)

Millions

of multiplications can go into a single prediction, too many to trace by hand.

Learned features
Which features did the network learn?

Saliency maps
How did each pixel contribute to a prediction?

Concepts
Which concepts did the network learn?

Adversarial examples
How can we fool the network?

Influential instances
Which training point influenced the prediction most?

SYNTHESIS

Which approach serves which goal?

	Debugging and improving	Justification	Insights into the data
By design	●●● Inner workings are visible	●●● Faithfully explains predictions	●○○ Model must reflect the world
Local, agnostic	●●○ Zooms in on unusual predictions	●●○ CP faithful; SHAP/LIME riskier	●●○ Compare to reference data
Global, agnostic	●●● E.g. improve the model with LOFO	●●○ Which features matter overall	●●● PDP, PFI reveal mechanisms
Model-specific	●●● Direct insight into the model	●●● Helps justify the model	●○○ Needs a theoretical justification

●●● strong ●●○ partial ●○○ use with caution

SYNTHESIS

The lines between categories are blurry

1

Is logistic regression interpretable by design?

Odds ratios need post-processed coefficients; marginal effects are arguably post-hoc.

2

Are boosted trees a black box?

With max depth 1, boosted tree stumps look like a generalized additive model.

3

Are Shapley values model-agnostic?

In linear regression, coefficient × value minus the average effect is exactly the Shapley value.



KEY MESSAGE

Interpretability is a fuzzy concept

Embrace the fuzziness, don't get too attached to one approach, and mix approaches as you need.

Thank you · Questions?

Molnar, C. (2025). “Methods Overview.” Interpretable Machine Learning, 3rd ed.

01 **By design**

Strong for debugging and justification; watch out for Rashomon.

02 **Model-agnostic**

SIPA; model and method can be swapped independently.

03 **Model-specific**

Reveals what neural networks have learned.